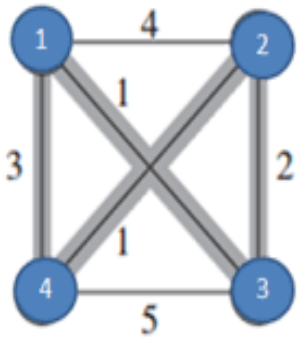
## Solve the Problem




| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0 | 4 | 1 | 3 |
| 2 | 4 | 0 | 2 | 1 |
| 3 | 1 | 2 | 0 | 5 |
| 4 | 3 | 1 | 5 | 0 |

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$g(1,\{2,3,4\})=\min\{c_{12}+g(2,\{3,4\}),c_{13}+g(3,\{2,4\}),c_{14}+g(4,\{2,3\})\}$

$g(2,\{3,4\}) = \min\{ c_{23} + g(3,\{4\}) , c_{24} + g(4,\{3\}) \}$

$g(3,\{4\})=c_{34}+c_{41}=5+3=8$

$g(4,\{3\})=c_{43}+c_{31}=5+1=6$

$g(2,\{3,4\})=\min\{ c_{23}+8,c_{24}+6\}=\min\{2+8,1+6\}=\min\{10,7\}=7$

$g(3,\{2,4\}) = \min\{ c_{32} + g\{2,\{4\}, c_{34} + g(4,\{2\}) ) \}$

$g(2,\{4\})= c_{24} + c_{41}=1+3=4$

$g(4,\{2\}) = c_{42} + c_{21}=1+4=5$

$g(3,\{2,4\}) = \min\{ c_{32} + 4, c_{34} + 5 \}=\min\{2+4,5+5\}$

$= \min\{6, 10\} = 6$

$g(4,\{2,3\})= \min\{ c_{42} + g\{2,\{3\}, c_{43} + g(3,\{2\}) \}$

$g\{2,\{3\}= c_{23} + c_{31}=2+1=3$

$g(3,\{2\})= c_{32}+c_{21}=2+4=6$

$g(4,\{2,3\})= \min\{ c_{42} + 3, c_{43} + 6 \}$

=min {1+3, 5+6} = min {4, 11} = 4

g (1,{2,3,4})=min {$c_{12}$+7,$c_{13}$+6,$c_{14}$+4}

=min {4+7, 1+6, 3+4}

= min {11, 7, 7} = 7

1, 3, 2, 4, 1=Optimal Path

Length=1+2+1+3=7

Optimal path length=7